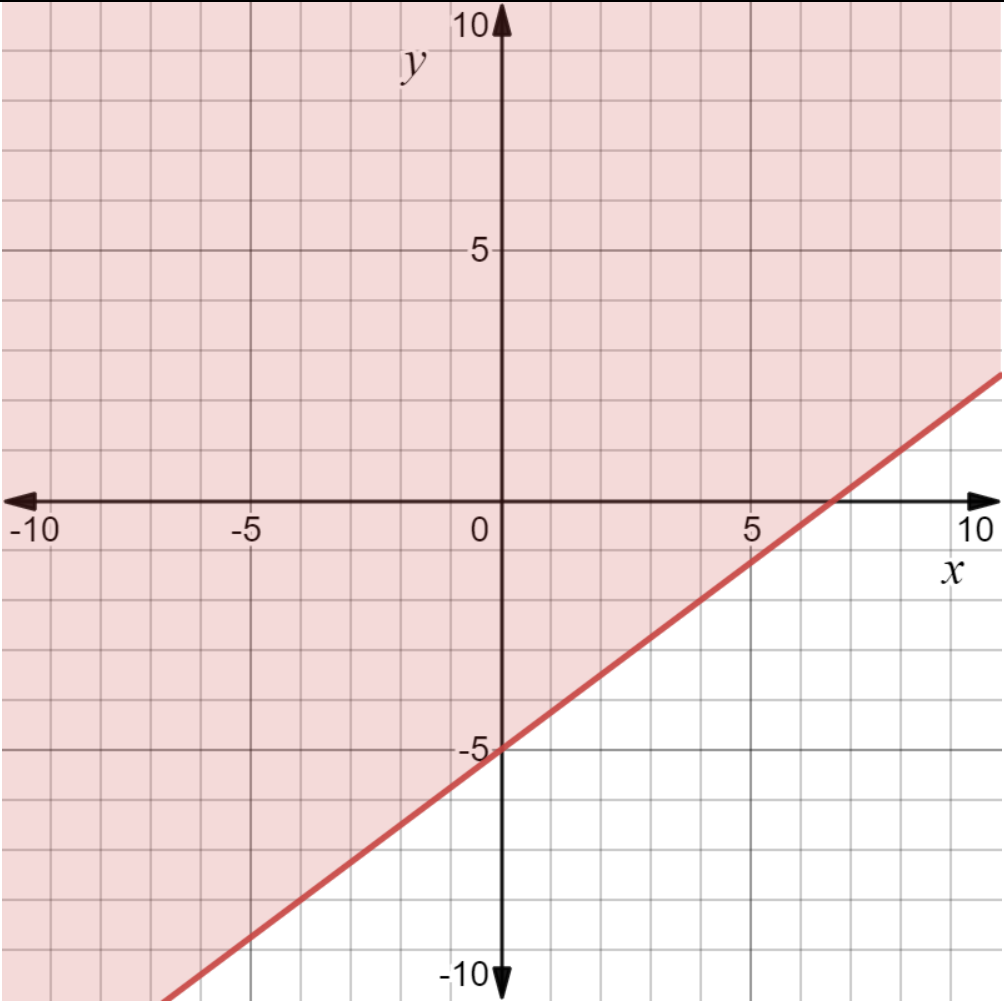


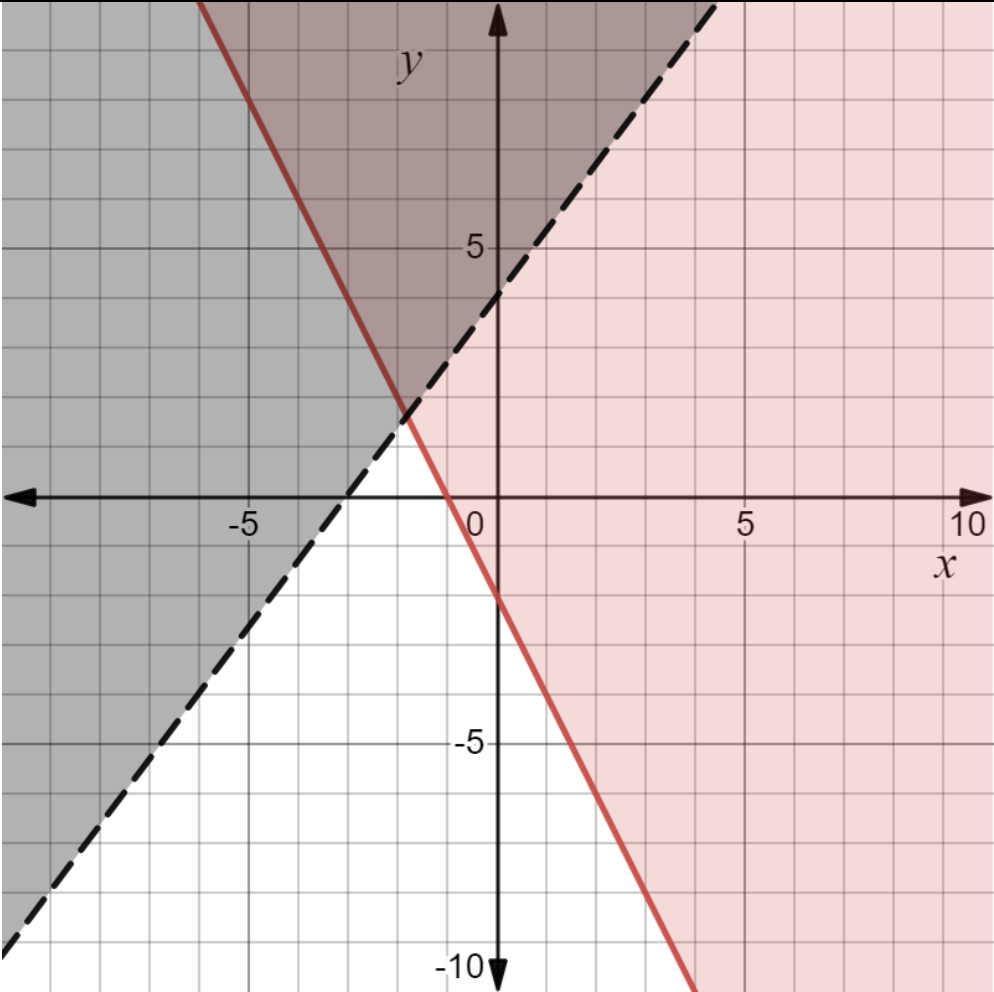
Algebra 1 Unit 6 Assessment Guide

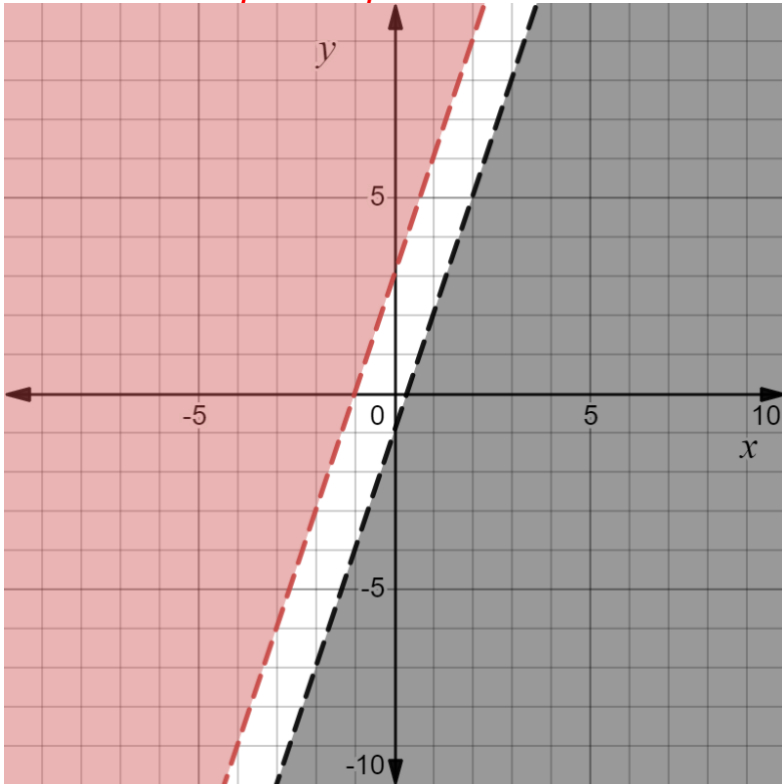
General Information	Writing inequalities for a real-world context is difficult for students. Students who understand how to write one equation can often transfer that understanding to writing an inequality or a system of inequalities. In grades 6-8, students have built a foundation of writing equations from a real-world context. In grade 6, students translated a real-world inequality. Students who struggle with writing inequalities for a real-world context may benefit from reviewing writing equations in a real-world context. Reviewing the following lessons from grades 6 and 7 may be helpful for students who struggle: 6 th Grade, Unit 11, Lesson 7 (real-world inequalities), 6 th Grade, Unit 12, Lesson 4, and 7 th Grade, Unit 10, Lessons 8-10. Students began solving inequalities in grade 7. Students who struggle with solving inequalities may benefit from reviewing the following lessons: 7 th Grade, Unit 10, Lessons 1-4 and 8 th Grade, Unit 5, Lessons 1-3. Sometimes students struggle with a context because they do not have the life experience. When remediating, help students “experience” the context as much as possible. Using manipulatives to explain the “story” of the context may help students learn to think about what in the context varies and needs to have a variable and what in the context stays constant.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.AR.2.6
	$x \leq 2.2$ 	Recommended Scoring (10 Total Points)	Consider 5 points for student work, 3 points for the correct answer, and 2 points for the correct number line graph. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether a minor deduction of points.
Question	2	Benchmark(s)	MA.912.AR.2.6
	$x < -3$ 	Recommended Scoring (10 Total Points)	Consider 5 points for student work, 3 points for the correct answer, and 2 points for the correct number line graph. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether a minor deduction of points.
Question	3	Benchmark(s)	MA.912.AR.2.6
	$-3 \leq x < 4$ 	Recommended Scoring (10 Total Points)	Consider 5 points for student work, 3 points for the correct answer, and 2 points for the correct number line graph. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether a minor deduction of points.

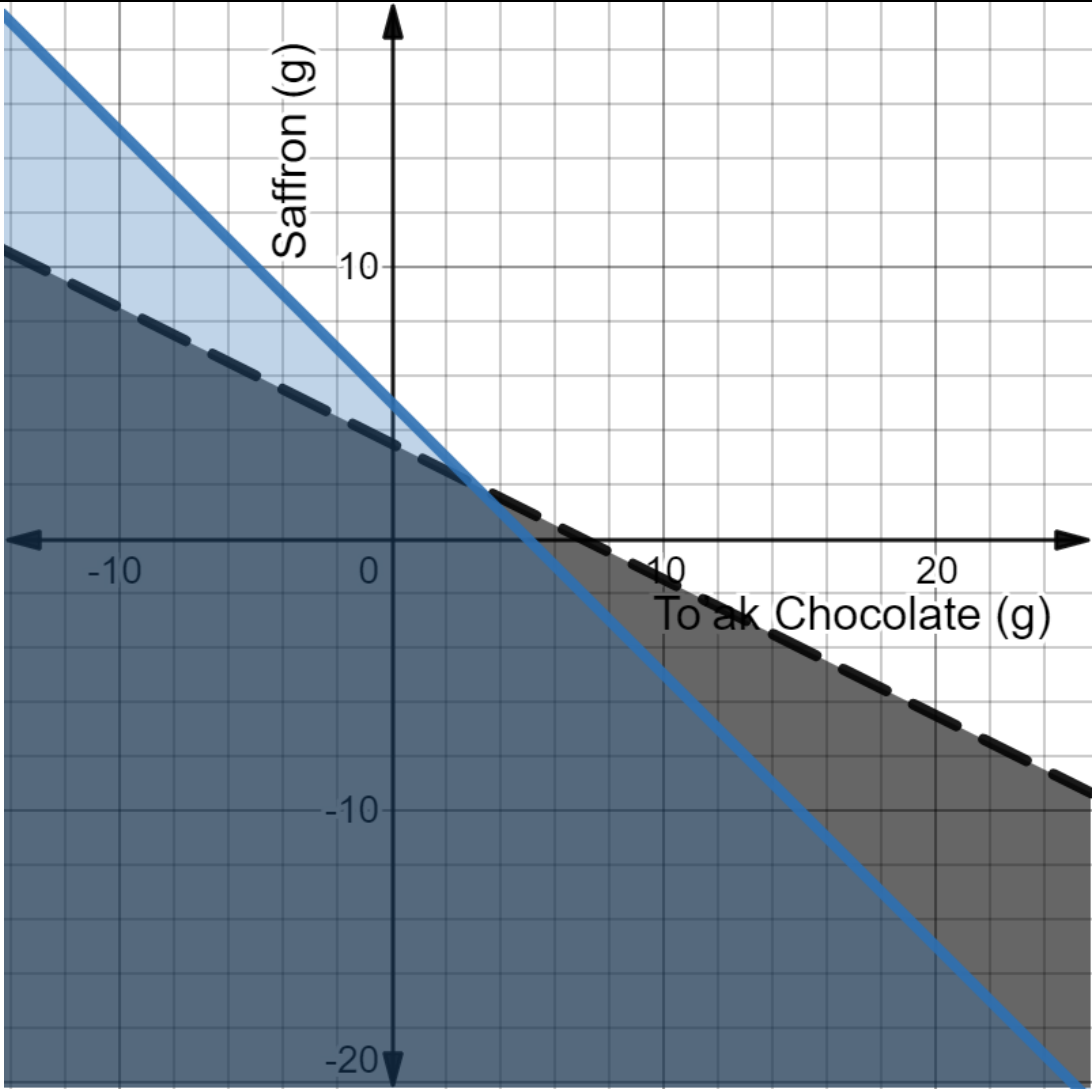
Question	4	Benchmark(s)	MA.912.AR.2.6
	Solution $115 \leq x \leq 495$	Recommended Scoring (10 Total Points)	Consider 7 points for student work and 3 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether a minor deduction of points.
Question	5	Benchmark(s)	MA.912.AR.2.6
	Inequality $20.9 \leq 0.68(x - 25) + 16.9 \leq 22$	Recommended Scoring (8 Total Points)	Consider giving students 2 points for the creating a compound inequality or two inequalities such that the expression is greater than or equal to 20.9 and less than or equal to 22. Consider giving students 2 points for using 0.68 as the slope in the expression. Consider giving students 2 points for the expression $(x - 25)$. Consider giving students 2 points for including 16.9 in the expression. Consider only deducting 2 points for students who write the expression $0.68x + 16.9$.

Question	6	Benchmark(s)	MA.912.AR.2.8
		Recommended Scoring (6 Total Points)	Consider 2 points for the correctly graphed line. Consider 1 point for using a solid line. Consider 3 points for the correct shading.

Question	7	Benchmark(s)	MA.912.AR.9.4
		Recommended Scoring (10 Total Points)	Consider 3 points for each correctly graphed line, including the line type. Consider deducting 1 point for each incorrect line type. Consider 4 points for the correct shading. Keep in mind that students can choose to just shade the area that is true for both inequalities.

Question	8	Benchmark(s)	MA.912.AR.9.4
<i>Answers will vary.</i> <i>A sample response is shown.</i> 		Recommended Scoring (8 Total Points) Any system where the lines are parallel and the shading does not overlap is correct. Partial credit is not recommended.	

Question	9	Benchmark(s)	MA.912.AR.9.1
System of Inequalities	$y \leq -5x + 9$ $y > \frac{4}{5}x - 2$	Recommended Scoring (10 Total Points)	Consider giving 3 points for each correct algebraic representation of each line (a total of 6 points). Consider that students may accidentally think a line crosses at different coordinate pairs so look for any evidence of this if a student's slope is close to the correct slope. Consider giving 2 points for each inequality that correctly matches the type of line (a total of 4 points).
Question	10a	Benchmark(s)	MA.912.AR.9.1
System of Inequalities	$x + y \leq 5$ $5x + 10y < 35$	Recommended Scoring (8 Total Points)	Consider giving 4 points for each correct algebraic representation of each inequality of the system. Consider that students may add $x \geq 0$ and $y \geq 0$. This is a more accurate representation of the real-world context. Students may also write two compound inequalities: $0 < x + y \leq 5$ $0 < 5x + 10y \leq 35$ This is also a more accurate representation of the real-world context.

Question	10b	Benchmark(s)	MA.912.AR.9.4
		Recommended Scoring (10 Total Points)	Consider following the student work for question 10. If a student made an error but correctly graphed the system, they created then the student should not lose the points for graphing. Consider 3 points for each correctly graphed line, including the line type. Consider deducting 1 point for each incorrect line type. Consider 4 points for the correct shading. Keep in mind that students can choose to just shade the area that is true for both inequalities